Capstone Executive Report

Business Goal:
The objective of this project is to accurately predict whether an individual earns >50K annually using demographic and employment data. A robust predictive model allows the business to better target marketing campaigns and optimize resource allocation for high-value customers.

Data Used:
We used the Adult Census Income dataset containing approximately 48,000 records. The dataset suffers from a significant class imbalance, with only ~24% of individuals falling into the >50K category. A robust preprocessing pipeline was applied to handle missing values, one-hot encode categorical variables, and scale numeric features.

Task 1: Ensemble Models & Task 2: Class Imbalance

We trained and compared Random Forest, HistGradientBoosting, and a Stacking Ensemble. To handle class imbalance, we evaluated Class Weighting, Random Under-Sampling, and SMOTE using 3-fold cross-validation.

Imbalance Strategy Performance (F1 Score):
- Class_Weight: CV = 0.6416, Holdout = 0.6429
- RandomUnderSampler: CV = 0.6463, Holdout = 0.6364
- SMOTE: CV = 0.6495, Holdout = 0.6419

Chosen Model & Deployment Artifact:
The Random Forest model with 'balanced' class weights provided the most stable F1-score across CV folds without the heavy computational overhead of SMOTE. This model was bundled with the preprocessing steps into 'final_model_pipeline.joblib'.

Task 3: Interpretability & Fairness Findings

Global Explanations:
Permutation Importance and SHAP values revealed the most influential features. The top drivers for predicting high income are capital-gain, education-num, age, marital-status, and hours-per-week. Local SHAP explanations successfully isolated why specific individuals were classified as True Positives or False Negatives.

Top Features (Permutation Importance):

1. education_hours
2. education-num
3. occupation_Exec-managerial
4. log_capital_gain
5. higher_education
6. hours_bucket_Part-Time
7. capital-gain
8. education_11th

Fairness Observations:
A check across protected groups (sex, race) shows disparities in the raw data, which the model reflects. Women and minority groups have lower >50K predictions due to historical biases present in the training set. Mitigations include threshold adjustments per group in production.

Recommended Next Steps & Monitoring

1. Deploy inference.py behind a REST API for real-time scoring.
2. A/B Test Plan: Route 10% of marketing traffic to the model's predictions and compare conversion rates against the baseline strategy.
3. Monitoring Strategy:
- Monitor Data Drift (e.g. shifts in age/education histograms).
- Monitor Label Distribution (alert if predicted >50K ratio exceeds 30%).
- Retrain model quarterly or when performance degrades.